

Transport Layer Security (TLS)

1. Introduction

Transport Layer Security (TLS) is a **cryptographic protocol** designed to provide **secure communication** over a computer network.

It ensures that data sent between a client (like a web browser) and a server (like a website) is **encrypted, authenticated, and protected** from tampering or eavesdropping.

- **TLS** is the **successor to SSL (Secure Sockets Layer)**.
- It operates between the **Application Layer** and **Transport Layer** in the OSI model.
- Used in protocols such as **HTTPS, FTPS, SMTP, IMAPS, POP3S** etc.

2. Objectives of TLS

TLS provides **three main security services**:

1. Confidentiality –

Ensures that data exchanged between sender and receiver cannot be read by unauthorized users.

(Achieved through **encryption**).

2. Integrity –

Ensures that the data is not altered or tampered with during transmission.

(Achieved through **message authentication codes – MACs**).

3. Authentication –

Confirms the identity of the communicating parties (client and server).

(Achieved through **digital certificates**).

3. How TLS Works (Process Overview)

TLS works through a process called the **TLS Handshake**, which happens when a secure connection is initiated.

Steps in TLS Handshake:

1. Client Hello:

- a. The client (e.g., browser) sends a message proposing encryption algorithms and a random number to the server.

2. Server Hello:

- a. The server responds with its chosen encryption algorithm and sends its **digital certificate** (contains public key).

3. Certificate Verification:

- a. The client verifies the server's certificate using a trusted Certificate Authority (CA).

4. Key Exchange:

- a. Both sides agree on a **session key** (temporary encryption key) using methods like **RSA** or **Diffie-Hellman**.

5. Finished Messages:

- a. Both the client and server confirm that future communication will be encrypted using the agreed session key.

6. Secure Communication:

- a. Encrypted and authenticated communication begins.

4. TLS Protocol Layers

TLS is divided into two main sub-protocols:

1. TLS Handshake Protocol

- a. Establishes the secure session, authentication, and encryption keys.

2. TLS Record Protocol

- a. Manages encryption, decryption, and data transmission after the handshake.

5. Advantages of TLS

- Provides end-to-end encryption
- Prevents eavesdropping and data theft
- Authenticates servers (and optionally clients)

6. Disadvantages / Limitations

- Slightly increases CPU and network load (due to encryption)
- Requires valid certificates and configuration
- Older versions (TLS 1.0/1.1) are vulnerable and deprecated

7. Real-Life Examples of TLS Use

- **HTTPS websites:** All secure websites use TLS (e.g., <https://google.com>).
- **Email security:** Used in SMTP over TLS (SMTPS).
- **VPN connections:** Many VPNs use TLS for encryption.
- **Mobile apps:** Secure API communication through TLS.